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Kinetic analysis and gas–liquid balances of the production of fermentative aromas during winemaking fermentations: Effect of assimilable nitrogen and temperature

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ABSTRACT

The impact of temperature and assimilable nitrogen on synthesis of the principal fermentative aromas by yeasts was evaluated with an online GC system. Kinetic parameters and gas–liquid balances were precisely calculated, making it possible to differentiate between the biological and physical consequences of temperature shifts. The lower liquid concentrations of esters at high temperature resulted mostly from large losses by evaporation, with only limited changes in yeast metabolism. Nitrogen availability was the main fermentation parameter affecting the kinetics of higher alcohol and ester synthesis, but the magnitude of this effect depended on temperature. The production of fermentative aromas was closely linked to central carbon metabolism, except for propanol. Indeed, two successive linear production phases (with yields dependent on initial nitrogen content and temperature) were obtained when volatile compound synthesis was plotted against sugar consumption. Overall and specific rates of isobutanol and isoamyl alcohol production were affected differently by nitrogen and temperature, indicating major differences in the regulation of synthesis for these two compounds, despite their partially shared metabolic pathways. Finally, propanol was produced exclusively during the nitrogen consumption phase, and the amount produced increased with initial nitrogen concentration. This compound was, therefore, identified as a quantitative, metabolic marker of the availability of intracellular nitrogen.

1. Introduction

Wine aroma is one of the principal attributes determining the preferences of wine consumers (Lee & Noble, 2003; Swiegers, Bartowsky, Henschke, & Pretorius, 2005; Ugliano, Travis, Francis, & Henschke, 2010). However, it is difficult to evaluate, because several hundred compounds can contribute to wine flavor (Francis & Newton, 2005; Swiegers et al., 2005). Indeed, wine aroma depends on a mixture of volatile compounds originating from the grapes used (varietal aromas), secondary metabolites synthesized by microorganisms during the winemaking fermentation (fermentative aromas) and molecules produced during aging (post-fermentative aromas). In addition, these volatile compounds can interact with each other and with constituents of the matrix (Escudero, Campo, Farina, Cacho, & Ferreira, 2007; Ferreira, Lopez, & Cacho, 2000). Most fruity aroma compounds, including esters in particular, are secondary metabolites produced by yeasts during alcoholic fermentation (Ugliano et al., 2010). Several studies have assessed the influence of fermentation parameters on the final fermentative aroma compound content of the wine, focusing mostly on higher alcohols and esters (Swiegers et al., 2005).

The concentration of assimilable nitrogen has a major influence on final fermentative aroma content (Swiegers et al., 2005). Initial nitrogen and final higher alcohol concentrations are directly correlated when assimilable nitrogen content is low. Conversely, an inverse relationship between initial nitrogen and final higher alcohol contents is observed in musts containing moderate to high levels of nitrogen (Carrau et al., 2008; Vilanova, Siebert, Varela, Pretorius, & Henschke, 2012; Vilanova et al., 2007). The final liquid concentrations of both acetate and ethyl esters are generally higher for musts with high nitrogen contents (Garde-Cerdan & Ancin Azpilicueta, 2008; Hernandez-Orte, Bely, Cacho, & Ferreira, 2006; Torrea et al., 2011; Ugliano et al., 2010). However, depending on the strain, the form of nitrogen added and the timing of nitrogen addition, lower levels of ester production may also be observed after nitrogen supplementation (Beltran, Esteve-Zarzoso, Rozes, Mas, & Guillamon, 2005; Jimenez-Marti, Aranda, Mendes-Ferreira, Mendes-Faia, & li del Olmo, 2007). *Saccharomyces cerevisiae* generally produces larger amounts of higher alcohols at high temperature (Beltran, Novo, Guillamon, Mas, & Rozes, 2008; Gamero, Tronchoni, Querol, & Belloch, 2013), but this relationship is not systematic (Molina, Swiegers, Varela, Pretorius, & Agosin, 2007). Final liquid

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concentrations of esters are systematically lower at high temperature than at lower temperatures, particularly for ethyl esters (Beltran et al., 2008; Molina et al., 2007). Nevertheless, much remains unknown about the way in which fermentation parameters regulate aroma metabolism.

In this study, we carried out online monitoring of the main fermentative aromas, to increase our understanding of yeast metabolism. The high acquisition frequency of online gas chromatography (GC) makes it possible to calculate the overall and specific rates of synthesis for volatile compounds (Mouret et al., 2013). We also determined the gas–liquid balances of aroma production (Morakul et al., 2013) to take losses into account and to distinguish between microbial and physicochemical phenomena.

2. Materials and methods

2.1. Fermentation

2.1.1. Yeast strain and culture media

Fermentations were carried out with the commercial *S. cerevisiae* strain Lalvin EC 1118® (Lallemand SA, Montreal, Canada). Fermentation tanks were inoculated with 200 mg l^{-1} active dry yeast that had been rehydrated by incubation with 50 g l^{-1} glucose for 30 min at 35 °C.

We used three different synthetic media (SMn, with 'n' indicating the assimilable nitrogen content) mimicking grape musts (Bely, Sablayrolles, & Barre, 1990). The base medium contained 90 g l^{-1} glucose, 90 g l^{-1} fructose, 6 g l^{-1} malic acid, 6 g l^{-1} citric acid, salts (0.75 g l^{-1} KH_2PO_4; 0.50 g l^{-1} K_2SO_4; 0.25 g l^{-1} $MgSO_4$; 0.155 g l^{-1} $CaCl_2$; 0.20 g l^{-1} NaCl), vitamins (20 mg l^{-1} myo-inositol; 1.5 mg l^{-1} pantothenic acid; 0.25 mg l^{-1} thiamine; 2 mg l^{-1} nicotinic acid, 0.25 mg l^{-1} pyridoxine; 0.003 mg l^{-1} biotin), trace elements (4 mg l^{-1} $MnSO_4$; 4 mg l^{-1} $ZnSO_4$; 1 mg l^{-1} $CuSO_4$; 1 mg l^{-1} KI; 0.4 mg l^{-1} $CoCl_2$; 1 mg l^{-1} H_3BO_3; 1 mg l^{-1} $(NH_4)_6Mo_7O_{24}$) and anaerobic factors (0.05% v/v Tween 80; 15 mg l^{-1} ergosterol; 0.0005% v/v oleic acid). Tartaric acid was replaced with citric acid to prevent the formation of tartrate precipitate during freezing at −20 °C. The source of nitrogen was a mixture of ammonium (30%) and amino acids (70%). Assimilable nitrogen concentration ranged from 70 to 410 mg N l^{-1} (SM70, SM230 and SM410), but the proportions of the different sources were identical in all three media. The amino-acid content of the medium was as follows (quantities as for SM410): 42.9 mg l^{-1} aspartic acid, 118.7 mg l^{-1} glutamic acid, 140.2 mg l^{-1} alanine, 361.2 mg l^{-1} arginine, 12.6 mg l^{-1} cysteine, 487.5 mg l^{-1} glutamine, 17.7 mg l^{-1} glycine, 31.5 mg l^{-1} histidine, 31.5 mg l^{-1} isoleucine, 46.7 mg l^{-1} leucine, 16.4 mg l^{-1} lysine, 30.3 mg l^{-1} methionine, 36.7 mg l^{-1} phenylalanine, 591.0 mg l^{-1} proline, 75.7 mg l^{-1} serine, 73.2 mg l^{-1} threonine, 173.0 mg l^{-1} tryptophan, 17.7 mg l^{-1} tyrosine and 42.9 mg l^{-1} valine.

2.1.2. Assimilable nitrogen determination

Ammonium concentration was determined enzymatically (R-Biopharm, Darmstadt, Germany).

The free amino-acid content of the must was determined by cation exchange chromatography, with post-column ninhydrin derivatization (Biochrom 30, Biochrom, Cambridge, UK) as described by Crepin, Nidelet, Sanchez, Dequin, and Camarasa (2012).

2.1.3. Cell counts

During fermentation, the cell population was determined with a Coulter Counter (model Z2, Beckman-Coulter, Margency, France) fitted with a 100 μm-aperture probe.

2.1.4. Tanks and fermentation control

Fermentations were run at pilot scale, in 10-liter stainless steel tanks, at 18, 24 and 30 °C. The amount of CO_2 released was measured accurately and automatically with a gas mass flow meter, for calculation of the rate of CO_2 production (dCO_2/dt).

As individual experiments were quite cumbersome to perform, each fermentation was performed only once. However, it has been shown that duplicate experiments run with this online monitoring system yield highly reproducible results. For all the various volatile compounds assessed, the relative standard deviation between duplicates was very low throughout the fermentation process (Mouret, Morakul, Nicolle, Athes, & Sablayrolles, 2012; Mouret et al., 2013): 4%, 3%, 6%, 3%, 2% and 4% for propanol, isobutanol, isoamyl alcohol, isoamyl acetate, ethyl hexanoate and ethyl octanoate, respectively.

2.2. Analysis of higher alcohols and esters

The concentrations of volatile compounds in the headspace of the fermentor were measured with an online GC device. The concentrations in the liquid phase were calculated as described by Morakul et al. (2011) and Morakul et al. (2013). Some measurements of concentration in the liquid phase were also performed, to validate estimation by the model. The relative standard deviation between model-based estimates and experimental determined values was 3%, validating the use of the model for this purpose.

2.2.1. Online measurements of concentrations in the gas phase

Headspace gas was pumped from the tank at a flow rate of 14 ml min^{-1}, through a heated transfer line. Carbon compounds were concentrated in a cold trap (Tenax TM) for 6 min, desorbed at 160 °C for 1 min, and analyzed with a Perichrom PR2100 GC coupled to a flame ionization detector (Alpha MOS, Toulouse, France). The details of the GC method and the calibration procedure were as previously described by Mouret et al. (2013).

2.2.2. Offline measurements of the liquid

Liquid samples were collected at various stages of the fermentation. Samples were centrifuged at 1500 ×g at 4 °C for 10 min and the supernatants were collected and stored at −20 °C until analysis. Sample preparation and the GC method used for the analysis were as described by Morakul et al. (2011, 2013) and Mouret et al. (2012).

2.3. Volatile compound balances during fermentation

2.3.1. Concentration in the liquid

The concentration of a volatile compound in the liquid ($C^{liq}(t)$) was calculated from the concentration measured online in the gas phase, expressed as $C^{gas}(t)$ in mg l^{-1} of CO_2, using the partition coefficient (k_i) value (Eq. (1)).

$$C^{liq}(t) = \frac{C^{gas}(t)}{k_i}. \tag{1}$$

The value of k_i (Eq. (2)) was calculated with the model developed by Morakul et al. (2011), as a function of the fermenting must composition, characterized by ethanol concentration and temperature.

$$\ln k_i = \mathrm{F1} + \mathrm{F2} \times E - \frac{\mathrm{F3} + \mathrm{F4} \times E}{R}\left(\frac{1000}{T} - \frac{1000}{T_{ref}}\right) \tag{2}$$

where E is the ethanol concentration (g l^{-1}) in the liquid phase, calculated from measurements of the amount of CO_2 released, which is proportional to sugar consumption; T is the current absolute temperature, T_{ref} corresponds to the absolute reference temperature (i.e. 293.15 K, 20 °C, in this study). F1, F2, F3 and F4 are constants, identified for each volatile compound. The values of these parameters for the different molecules were determined by Mouret et al. (2013). Ethyl octanoate was recently calibrated on the online GC system. So, for this particular compound, changes in the value of k_i with temperature and ethanol concentration were modeled from the data collected during the nine experiments presented in this work. For this particular compound, the numerical values of F1, F2, F3 and F4 were -3.13 ± 0.24, $-1.35 \times 10^{-2} \pm 4.3 \times 10^{-3}$, 52 ± 35, and $-3.60 \times 10^{-3} \pm 5.91 \times 10^{-1}$, respectively.

2.3.2. Losses in the exhaust gas

The losses in the gas were calculated with Eq. (3).

$$L(t) = \int_0^t C^{gas}(t) \times Q(t) \times dt \qquad 3$$

where Q(t) is the CO_2 flow rate at time t, expressed in liters of CO_2 per liter of must and per hour. The relative losses (RL), expressed as a percentage of production ($P(t)$), were determined as follows (Eq. (4)):

$$RL = \frac{L(t)}{P(t)} = \frac{\int_0^{t_{end}} C^{gas}(t) \times Q(t) \times dt}{C^{liq}(t_{end}) + \int_0^{t_{end}} C^{gas}(t) \times Q(t) \times dt} \qquad 4$$

where t_{end} corresponds to the end of fermentation (h).

2.3.3. Total production

The total production of a volatile compound at time t, expressed as $P(t)$ in mg l^{-1} of must, was calculated by adding the concentration in the liquid phase, expressed as $C^{liq}(t)$ in mg l^{-1} of must, to the amount of the volatile compound lost in the gas phase, expressed as $L(t)$ in mg l^{-1} of must (Eq. (5)).

$$P(t) = C^{liq}(t) + L(t). \qquad 5$$

This production reflects the ability of the yeast to produce a volatile compound, regardless of its subsequent fate: accumulation in the liquid phase or evaporation.

2.3.4. Calculation of the rates and specific rates of production, accumulation in the liquid and loss in the gas phase

The high frequency of online GC measurements (up to one measurement per hour) made it possible to calculate the rates of volatile compound production, accumulation and loss, using sliding-window second-order polynomial fitting in a custom-developed Labview application.

The specific rates were calculated by dividing the corresponding rates by the cell count.

2.4. Statistical analysis

R software version 1.14.2 was used for statistical analysis. FactoMineR package (Le, Josse, & Husson, 2008) was used for principal component analysis (PCA), for studies of the kinetics of fermentative aroma production.

3. Results and discussion

We investigated the impact of initial nitrogen concentration and temperature on the production of the principal fermentative aromas, performing nine fermentations at different temperatures and with different initial assimilable nitrogen concentrations. Six fermentative aromas from various chemical families were monitored throughout the fermentation process: three higher alcohols (propanol, isobutanol and isoamyl alcohol), one acetate ester (isoamyl acetate) and two ethyl esters (ethyl hexanoate and ethyl octanoate).

For all fermentations, nitrogen was exhausted at the end of the growth phase and the sugar supply was exhausted at the end of the culture.

3.1. Effect of nitrogen and temperature on the kinetics of volatile compound synthesis

We present here data for the total production of fermentative aromas (corresponding to the sum of liquid content and gaseous losses), because this reflects the true capacity of the yeast to synthesize these volatile compounds.

3.1.1. Higher alcohols

3.1.1.1. Propanol. The pattern of production for propanol was different from that of the other volatile molecules studied, as propanol production stopped at the end of the growth phase (Table 1), when cell count was maximal (Fig. 1a, b, c).

In medium with an initial nitrogen concentration of 410 mg N l^{-1}, nitrogen consumption, growth and propanol synthesis all ended at the same time (Fig. 1c). By contrast, in media containing 230 or 70 mg N l^{-1} assimilable nitrogen, cell count and propanol content continued to increase for a short period after exhaustion of external nitrogen supply (Fig. 1a, b). This suggests that the intracellular nitrogen pool may be managed differently in media with different initial nitrogen contents, with probable transient nitrogen storage and the subsequent consumption of these reserves when nitrogen is not provided in excess. Nevertheless, whatever the culture medium, cell count and propanol synthesis were maximal when the intracellular nitrogen supply was exhausted. Propanol can therefore be considered to be a metabolic marker of the availability of intracellular nitrogen. This observation is consistent with propanol being generated exclusively as a consequence of nitrogen metabolism, by the direct deamination of exogenous threonine or by interconversion between other amino acids and threonine, catalyzed by aminotransferases and transaminases (Boulton, Singleton, Bisson, & Kunkee, 1995).

A connection between nitrogen metabolism and propanol synthesis was also evident from the relationship between the rate of total propanol production and the consumption of its amino-acid precursor, threonine. The rate of total propanol production gradually increased during fermentation until the supply of threonine was exhausted, and declined thereafter (Fig. 2a, b). This observation suggests that propanol

Table 1
Percentage production of volatile compounds during the growth phase.[a]

Initial nitrogen concentration (mg l^{-1})	Fermentation temperature (°C)	Propanol	Isobutanol	Isoamyl alcohol	Isoamyl acetate	Ethyl hexanoate	Ethyl octanoate
70	18	80	55	24	20	11	31
70	24	87	46	24	17	12	15
70	30	81	24	14	9	6	5
230	18	97	26	19	16	25	22
230	24	89	27	15	13	22	18
230	30	97	19	14	12	24	17
410	18	83	37	37	23	46	36
410	24	79	33	28	17	41	31
410	30	79	32	31	17	37	29

[a] % when cell count reached 90% of its maximum value.

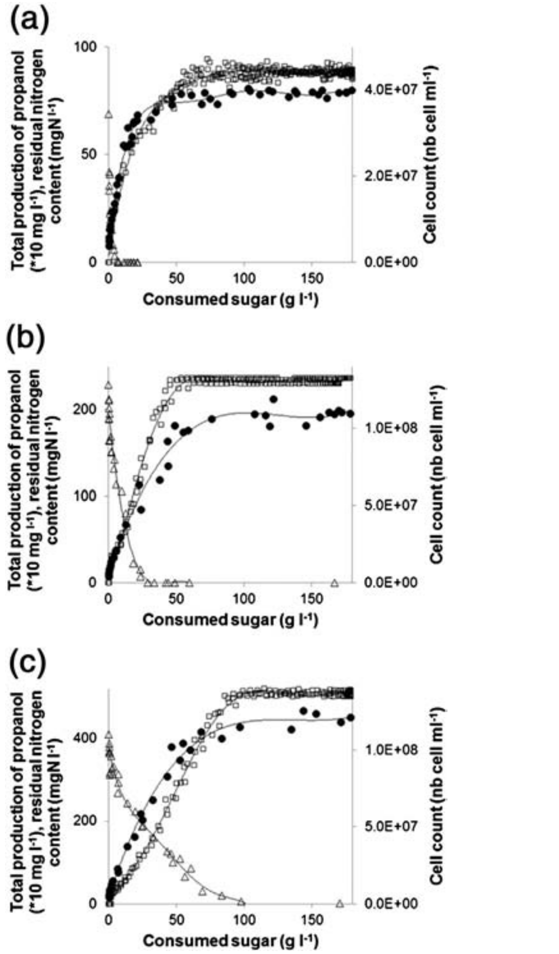

Fig. 1. Changes in the total production of propanol (□), cell count (●) and residual nitrogen content (Δ) for SM70 (a), SM230 (b) and SM410 (c), as a function of sugar consumption. The values shown on each graph correspond to three fermentations, performed at 18, 24 and 30 °C.

may act as a "safety valve", preventing the accumulation of α-ketobutyrate and threonine. Furthermore, the rate of total production of this higher alcohol followed the same pattern regardless of temperature (Fig. 2a), whereas major differences were observed between fermentations in media with different initial nitrogen contents (Fig. 2b).

Plots of total propanol production against assimilable nitrogen consumption highlighted two successive linear production phases (Fig. 3). The yield of the second phase was systematically greater than that of the first phase. These yields were independent of temperature (Fig. 3). The transition between the two phases was not related to exhaustion of the supply of threonine, the amino-acid precursor of propanol, but was almost certainly linked to the sequential use of different nitrogen sources (Crepin et al., 2012). Indeed, the transition coincided with the exhaustion of the nitrogen compounds used early in fermentation (Asp, Thr, Glu, Leu, His, Met, Ile, Ser, Gln, and Phe) and the start of consumption of the nitrogen compounds used late in fermentation (ammonium, Val, Arg, Ala, Trp, and Tyr).

Consistent with our observations concerning the kinetics of propanol production, total production levels throughout the fermentation process for this higher alcohol were independent of temperature and directly proportional to initial nitrogen content (Table 2). In all nine fermentations, final propanol concentration was linearly correlated with initial assimilable nitrogen content (regression coefficient = 0.9807). Thus, propanol synthesis can be used as a quantitative marker of initial assimilable nitrogen content. This proportionality between nitrogen concentration and propanol synthesis has never been described, although several studies (Clement et al., 2013; Gonzalez-Marco, Jimenez-Moreno, & Ancin-Azpilicueta, 2010; Mendes-Ferreira, Barbosa, Falco, Leao, & Mendes-Faia, 2009; Moreira, de Pinho, Santos, & Vasconcelos, 2011; Torrea, Fraile, Garde, & Ancin, 2003; Vilanova et al., 2007) report final propanol concentrations to be higher following the fermentation of musts with high assimilable nitrogen concentrations.

Propanol can therefore be considered to be both a metabolic marker of the availability of intracellular nitrogen and a quantitative marker of assimilable nitrogen content.

3.1.1.2. Isobutanol and isoamyl alcohol. Both isobutanol and isoamyl alcohol were produced principally during stationary phase (Table 1),

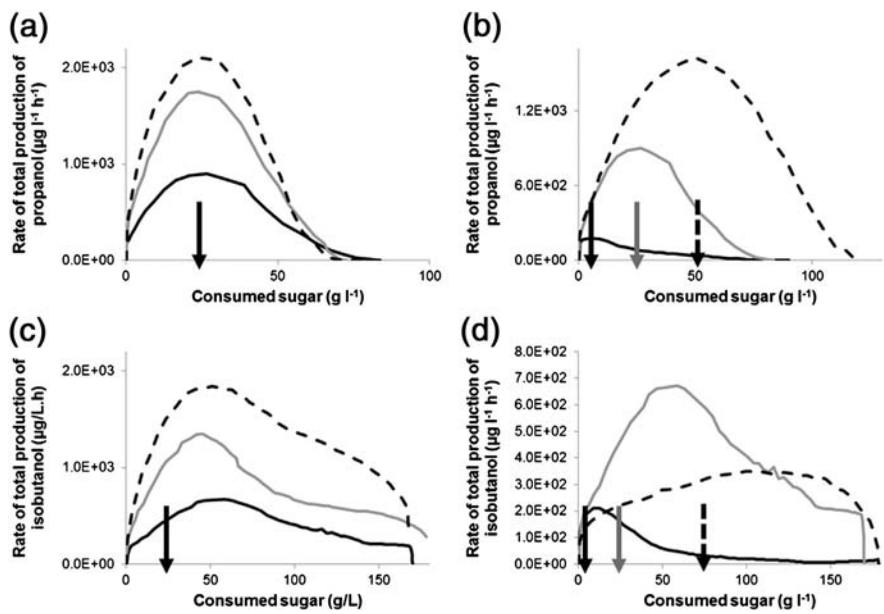

Fig. 2. (a) and (b) Changes in the rate of total propanol production at 18 °C (—), 24 °C (—) and 30 °C (— —) for an initial nitrogen concentration of 230 mg N l^{-1} (a) and for SM70 (—), SM230 (—) and SM410 (— —) at 18 °C (b). The arrows indicate the exhaustion of extracellular threonine for an initial nitrogen concentration of 230 mg N l^{-1} (a) and for SM70 (black), SM230 (gray) and SM410 (dotted) (b). (c) and (d) Changes in the rate of total isobutanol production at 18 °C (—), 24 °C (—) and 30 °C (— —) for an initial nitrogen concentration of 230 mg N l^{-1} (c) and for SM70 (—), SM230 (—) and SM410 (— —) at 18 °C (d). The arrows correspond to the exhaustion of extracellular threonine for an initial nitrogen concentration of 230 mg N l^{-1} (c) and for SM70 (black), SM230 (gray) and SM410 (dotted) (d).

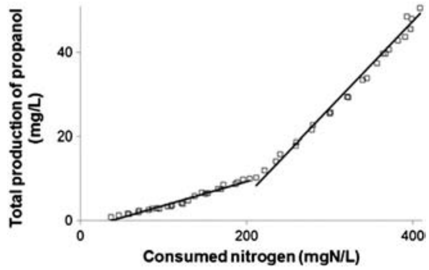

Fig. 3. Changes in total propanol production as a function of nitrogen consumption for an initial assimilable nitrogen content of 410 mg N l^{-1}, at the three different temperatures. For the two successive linear phases, the regression coefficients were 0.988 and 0.987, respectively.

regardless of experimental conditions, as previously reported by Mouret et al. (2013). A comparison of the production profiles of these aromatic molecules with the assimilation of their corresponding amino-acid precursors revealed no clear link between the synthesis of these higher alcohols and the consumption of their precursors. Indeed, as shown in Fig. 2c and d for isobutanol and valine, the amino-acid precursor was systematically exhausted before the rate of total production for the corresponding higher alcohol peaked. These two observations suggest that isobutanol and isoamyl alcohol are not produced exclusively from their amino-acid precursors (via the Ehrlich pathway) and that other metabolic pathways (such as the production of keto-acids from central carbon metabolism) make a significant contribution to their synthesis.

We also analyzed the relationship between the synthesis of isobutanol and isoamyl alcohol and sugar consumption. Two consecutive linear production phases were identified, as shown in Fig. 4a for isoamyl alcohol. The transition between the two linear phases of production from sugar was simultaneous for these two higher alcohols, corresponding in all cases to the end of the growth phase. This can be explained by the different metabolic pathways for these two higher alcohols, one being produced by the catabolism of exogenous amino acids and the other by central carbon metabolism (Swiegers et al., 2005; Van Der Sluis et al., 2002). These higher alcohols were generated by both carbon and nitrogen metabolism during the growth phase, whereas they were produced solely by central carbon metabolism during the stationary phase.

The production yield of the higher alcohols during the second phase was higher than that for the first phase at high nitrogen contents (230 and 410 mg N l^{-1}), as shown in Table 3a. By contrast, for medium with an initial nitrogen content of 70 mg l^{-1}, the yield of the second phase was lower. These differences in production yields suggest that (i) the contributions of central carbon and nitrogen metabolism to the synthesis of keto-acid precursors and their further conversion to isobutanol and isoamyl alcohol differ at high and low nitrogen contents and (ii) the regulation of the metabolic pathways involved in the production and consumption of keto-acids differ according to the pool of extracellular amino acids used.

For the three temperatures tested, both the production yields of these higher alcohols from sugars (Table 3a) and their final levels (Table 2) were maximal for an initial nitrogen content of 230 mg N l^{-1}. This indicates that the synthesis of higher alcohols is not a monotonic function of initial nitrogen content, with optimal production for an initial assimilable nitrogen concentration consistent with previous reports, at 200 to 300 mg l^{-1} (Carrau et al., 2008; Jimenez-Marti et al., 2007; Vilanova et al., 2012). Differences in the management of the keto-acid pools as a function of the quality and quantity of extracellular amino acids may explain these observations. Indeed, the intracellular availability of keto-acids is dependent on several different metabolic pathways: (1) production from central carbon metabolism, (2) the deamination of amino acids, (3) re-use for amino-acid synthesis (anabolic use) and (4) conversion into higher alcohols.

Temperature had different effects on the production of isobutanol and that of isoamyl alcohol, whatever the culture medium. For isobutanol, production yields and total production increased with temperature, whereas these parameters peaked at 24 °C for isoamyl alcohol (Tables 2 and 3a). This suggests that, surprisingly, temperature had different effects (metabolic and/or genomic effects) on the synthesis pathways of these two higher alcohols, despite their similarity, both these pathways including α-ketoisovalerate as an intermediate (Hazelwood, Daran, van Maris, Pronk, & Dickinson, 2008). These differences in the metabolic and/or genomic regulation of the synthesis of these two

Table 2
Total production, final concentration in the liquid phase, and relative losses of volatile compounds.

Initial nitrogen concentration (mg l^{-1})	Fermentation temperature (°C)	Propanol			Isobutanol			Isoamyl alcohol		
		Total production (mg l^{-1})	Final concentration in wine (mg l^{-1})	Relative losses (%)	Total production (mg l^{-1})	Final concentration in wine (mg l^{-1})	Relative losses (%)	Total production (mg l^{-1})	Final concentration in wine (mg l^{-1})	Relative losses (%)
70	18	8.82	8.78	0.45	17.1	17.0	0.58	167	166	0.60
70	24	8.80	8.75	0.57	35.7	35.4	0.98	231	228	1.30
70	30	10.0	9.94	0.60	66.8	66.0	1.20	203	200	1.48
230	18	23.1	23.0	0.43	35.8	35.6	0.56	205	204	0.49
230	24	25.6	25.4	0.78	53.7	53.3	0.74	244	242	0.82
230	30	24.1	23.8	1.24	82.9	82.1	0.97	226	223	1.33
410	18	51.5	51.3	0.39	25.8	25.7	0.39	120	119	0.83
410	24	50.5	50.2	0.59	37.4	37.2	0.53	138	136	1.46
410	30	50.5	50.0	0.99	39.7	39.3	1.01	111	109	1.80

Initial nitrogen concentration (mg l^{-1})	Fermentation temperature (°C)	Isoamyl acetate			Ethyl hexanoate			Ethyl octanoate		
		Total production (mg l^{-1})	Final concentration in wine (mg l^{-1})	Relative losses (%)	Total production (mg l^{-1})	Final concentration in wine (mg l^{-1})	Relative losses (%)	Total production (mg l^{-1})	Final concentration in wine (mg l^{-1})	Relative losses (%)
70	18	0.28	0.223	19	0.47	0.344	27	0.51	0.33	34
70	24	0.41	0.299	26	0.57	0.354	38	0.57	0.35	39
70	30	0.36	0.220	39	0.50	0.240	52	0.38	0.17	54
230	18	1.15	0.967	16	0.68	0.494	27	0.94	0.70	26
230	24	1.53	1.15	25	0.69	0.419	39	0.92	0.58	36
230	30	1.19	0.774	35	0.56	0.276	51	0.72	0.37	48
410	18	2.47	2.13	14	0.82	0.579	29	1.01	0.74	26
410	24	2.12	1.67	21	0.83	0.484	42	1.06	0.67	37
410	30	1.43	0.952	33	0.66	0.326	50	0.91	0.46	49

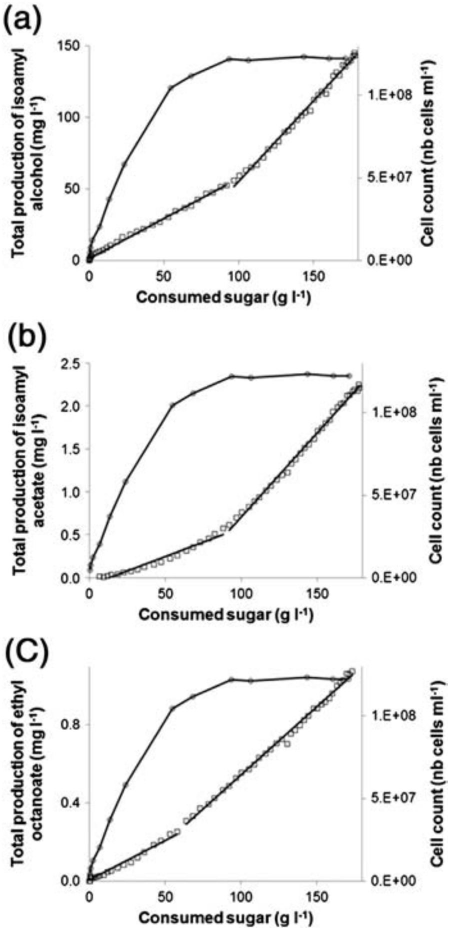

Fig. 4. (a) Changes in the total production of isoamyl alcohol (□) and cell count ( ) as a function of sugar consumption. For the two successive linear phases, the regression coefficients were 0.997 and 0.994, respectively. (b) Changes in the total production of isoamyl acetate (□) and cell count ( ) as a function of sugar consumption. For the two successive linear phases, the regression coefficients were 0.990 and 0.996, respectively. (c) Changes in the total production of ethyl octanoate (□) and cell count ( ) as a function of sugar consumption. For the two successive linear phases, the regression coefficients were 0.979 and 0.995, respectively. For all three graphs, the initial nitrogen concentration was 410 mg N l^{-1} and the fermentation temperature was 24 °C.

fermentative aroma compounds were confirmed by the very large differences in their rates and specific rates of total production (Fig. 5a, b).

3.1.2. Esters

3.1.2.1. Acetate ester. Like isoamyl alcohol, isoamyl acetate was produced mostly during the stationary phase (Table 1).

The phenomena observed depended on the initial nitrogen content. At high nitrogen contents (230 and 410 mg N l^{-1}), two consecutive linear production phases were observed when total acetate ester production was plotted against sugar consumption. The transition between the two phases occurred at the end of the growth phase (Fig. 4b). The yield of the second linear phase was much higher than that of the first (by 245% on average; Table 3a). The production yields and total production (Tables 2 and 3a) of this acetate ester increased with initial nitrogen concentration. A positive correlation between the synthesis of acetate esters and nitrogen availability has been reported in previous studies (Garde-Cerdan & Ancin Azpilicueta, 2008; Hernandez-Orte et al., 2006; Torrea et al., 2011; Ugliano et al., 2010) and is not surprising, because nitrogen metabolism, via the Ehrlich pathway, contributes to the formation of higher alcohols and their acetate ester derivatives (Swiegers et al., 2005). By contrast, there was only one linear production phase (Table 3a), in medium with an initial nitrogen content of 70 mg N l^{-1}, suggesting a minor contribution of nitrogen metabolism to aromatic molecule formation.

Like the relationship between total isoamyl acetate production and sugar consumption, the relationship between total isoamyl alcohol production and isoamyl acetate production was dependent on initial nitrogen content. At low nitrogen concentration (70 mg N l^{-1}), there were two consecutive linear production phases, the second having a higher yield than the first (Fig. 6b and Table 3b). By contrast, at high initial nitrogen contents (230 and 410 mg N l^{-1}), there was only one linear production phase, which began when isoamyl alcohol reached a threshold concentration (Fig. 6a and Table 3b). This threshold may correspond to an intracellular concentration of isoamyl alcohol required for the induction of ester synthesis. This hypothesis is supported by the much higher *Km* value of the alcohol acetyl transferase (EC number: 2.3.1.84) for isoamyl alcohol than for acetyl-coA (Minetoki, Bogaki, Iwamatsu, Fujii, & Hamachi, 1993): 29.8 and 0.025 mM for isoamyl alcohol and acetyl-coA, respectively. Due to this high *Km*, an intracellular accumulation of isoamyl alcohol is required to induce the production of the ester or to achieve a production rate high enough for the detection of extracellular isoamyl acetate. Alternatively, there may be an effect of enzyme activity or enzyme gene transcription. In medium with an initial nitrogen content of 70 mg N l^{-1}, the cell population was small and its anabolic demand for amino acids was low. The threshold level of isoamyl alcohol required to induce isoamyl acetate synthesis was therefore probably reached rapidly. This may account for the lack of a time lag in the production of this acetate ester from its precursor higher alcohol in medium with a low nitrogen content. However, it does not account for the two successive phases in the medium containing 70 mg N l^{-1} assimilable nitrogen. Other regulatory systems must therefore be involved.

All these observations indicate that the intracellular concentration of higher alcohols is a key element in the synthesis of acetate esters. Nevertheless, substrate availability is not the only factor. The activity of the alcohol acetyltransferases catalyzing the final step of acetate ester synthesis (Lilly, Lambrechts, & Pretorius, 2000) is also regulated at the level of transcription (Lilly et al., 2006; Saerens, Delvaux, Verstrepen, & Thevelein, 2010; Verstrepen, Van Laere, et al., 2003).

Finally, we noted that the ranking of production yields and total production according to temperature differed as a function of nitrogen availability (Tables 2 and 3a), indicating an interaction between the effects of initial nitrogen content and temperature on the overall production of this acetate ester.

3.1.2.2. Ethyl esters. Ethyl hexanoate and ethyl octanoate were produced mostly during the stationary phase (Table 1).

On a plot of the total production of these ethyl esters against sugar consumption, two consecutive linear production phases were observed, as shown in Fig. 4c for ethyl octanoate. For the two ethyl esters, whatever the initial nitrogen concentration, the transition between the two linear production phases occurred at the same point in the fermentation (after the consumption of 20 g l^{-1} and 60 g l^{-1} sugar for ethyl hexanoate and ethyl octanoate (Fig. 4c), respectively). Ethyl esters are produced by lipid metabolism; differences in yield were therefore not connected to nitrogen metabolism.

In all fermentation conditions (except for ethyl octanoate at 18 °C and an initial nitrogen concentration of 70 mg N l^{-1}), the yield of the second phase was higher than that of the first phase, by 297% and 132% for ethyl hexanoate and ethyl octanoate, respectively. Yields were maximal at 18 °C and 410 mg N l^{-1} assimilable nitrogen (Table 3a) for the two ethyl esters. Surprisingly, production yield was influenced by the initial nitrogen concentration. This may be due to connections between nitrogen and lipid metabolism via redox regulation or the possible synthesis of acyl-CoA derivatives from α-ketoacids. An alternative explanation would be the induction of acyl-coA synthesis due to the higher lipid requirements for high levels of biomass production in media with high nitrogen concentrations.

Table 3
Production yields of fermentative aromas during the two successive linear production phases.

(a) Yields from sugars										
Production yield from sugar (mg g^{-1})		SM70, 18 °C	SM70, 24 °C	SM70, 30 °C	SM230, 18 °C	SM230, 24 °C	SM230, 30 °C	SM410, 18 °C	SM410, 24 °C	SM410, 30 °C
Isobutanol	First phase	1.84×10^{-1}	3.51×10^{-1}	6.53×10^{-1}	1.69×10^{-1}	3.22×10^{-1}	3.41×10^{-1}	1.16×10^{-1}	1.79×10^{-1}	2.06×10^{-1}
	Second phase	3.50×10^{-2}	1.11×10^{-1}	2.64×10^{-1}	2.09×10^{-1}	2.94×10^{-1}	5.02×10^{-1}	1.98×10^{-1}	2.84×10^{-1}	2.50×10^{-1}
Isoamyl alcohol	First phase	1.17	1.51	1.71	6.74×10^{-1}	8.22×10^{-1}	6.81×10^{-1}	5.18×10^{-1}	5.60×10^{-1}	5.49×10^{-1}
	Second phase	7.65×10^{-1}	1.20	8.99×10^{-1}	1.53	1.75	1.60	9.71×10^{-1}	1.13	7.09×10^{-1}
Isoamyl acetate	First phase	1.16×10^{-3}	1.61×10^{-3}	1.21×10^{-3}	1.68×10^{-3}	2.95×10^{-3}	2.84×10^{-3}	4.62×10^{-3}	4.88×10^{-3}	2.99×10^{-3}
	Second phase				6.61×10^{-3}	7.89×10^{-3}	5.28×10^{-3}	1.85×10^{-1}	1.44×10^{-2}	7.22×10^{-3}
Ethyl hexanoate	First phase	8.44×10^{-4}	5.34×10^{-4}	7.69×10^{-4}	1.61×10^{-3}	8.21×10^{-4}	4.17×10^{-4}	9.39×10^{-4}	1.93×10^{-3}	1.79×10^{-3}
	Second phase	2.02×10^{-3}	2.47×10^{-3}	2.56×10^{-3}	2.96×10^{-3}	2.42×10^{-3}	1.92×10^{-3}	4.36×10^{-3}	4.01×10^{-3}	1.63×10^{-3}
Ethyl octanoate	First phase	3.11×10^{-3}	7.99×10^{-4}	4.79×10^{-4}	2.76×10^{-3}	2.06×10^{-3}	1.29×10^{-3}	2.73×10^{-3}	2.26×10^{-3}	2.17×10^{-3}
	Second phase	1.26×10^{-3}	2.17×10^{-3}	1.68×10^{-3}	3.92×10^{-3}	3.57×10^{-3}	2.51×10^{-3}	4.85×10^{-3}	3.91×10^{-3}	2.65×10^{-3}
(b) Yields of isoamyl acetate from its higher alcohol precursor										
Production yield from isoamyl alcohol (mg g^{-1})		SM70, 18 °C	SM70, 24 °C	SM70, 30 °C	SM230, 18 °C	SM230, 24 °C	SM230, 30 °C	SM410, 18 °C	SM410, 24 °C	SM410, 30 °C
Isoamyl acetate	First phase	1.16×10^{-3}	1.29×10^{-3}	1.18×10^{-3}	5.85×10^{-3}	6.38×10^{-3}	5.51×10^{-3}	2.45×10^{-2}	1.77×10^{-2}	1.58×10^{-2}
	Second phase	2.55×10^{-3}	2.51×10^{-3}	3.03×10^{-3}						

3.1.3. General overview

With the exception of propanol, the fermentative aromas were produced principally during the stationary phase (Table 1). This observation confirms the results obtained by Mouret et al. (2013) for a natural must and is consistent with fermentative aromas generally not being synthesized directly by the degradation of amino acids (Miller, Wolff, Bisson, & Ebeler, 2007).

Nevertheless, the percentage fermentative aroma production during the growth phase (estimated when the cell count reached 90% of its maximum value) varied with temperature and initial nitrogen content. For all the volatile compounds studied, percentage production during the growth phase decreased with temperature (Table 1). This observation suggests that the pools of keto-acids and fatty acids may be distributed differently between anabolism and volatile compound production as a function of temperature and, probably, growth rate. One key finding was that percentage ester production during the growth phase increased with initial nitrogen content, particularly for ethyl esters (Table 1). This was probably because the growth phase was longer for higher initial nitrogen concentrations. Indeed, when the initial nitrogen concentration was 70 mg N l^{-1}, the growth phase ended when 45 g sugar l^{-1} had been consumed, whereas, at an initial nitrogen concentration of 410 mg N l^{-1}, the growth phase continued until 100 g sugar l^{-1} had been consumed.

Moreover, there was a relationship between the total production of fermentative aromas and sugar consumption, for all the compounds considered other than propanol. For these volatile compounds, two consecutive linear production phases were generally identified. The durations of these two phases depended on the compound studied. The yields obtained during the two phases varied with initial nitrogen content and temperature (Table 3a).

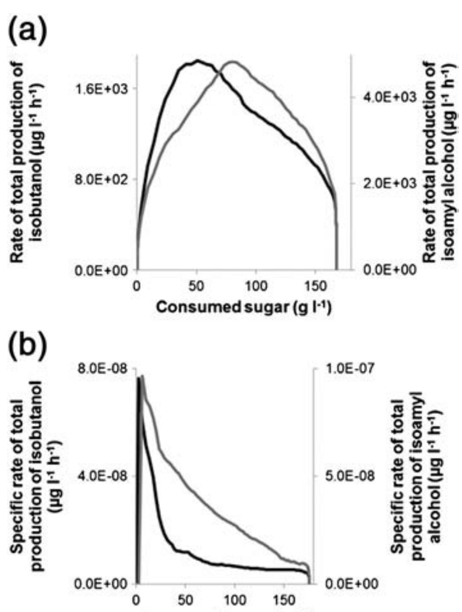

Fig. 5. (a) Changes in the rate of total production of isobutanol (—) and isoamyl alcohol (—) as a function of sugar consumption, for an initial assimilable nitrogen content of 230 mg N l^{-1} and a temperature of 30 °C. (b) Changes in the specific rates of total production for isobutanol (—) and isoamyl alcohol (—) as a function of sugar consumption, for an initial assimilable nitrogen content of 70 mg N l^{-1} and a temperature of 30 °C.

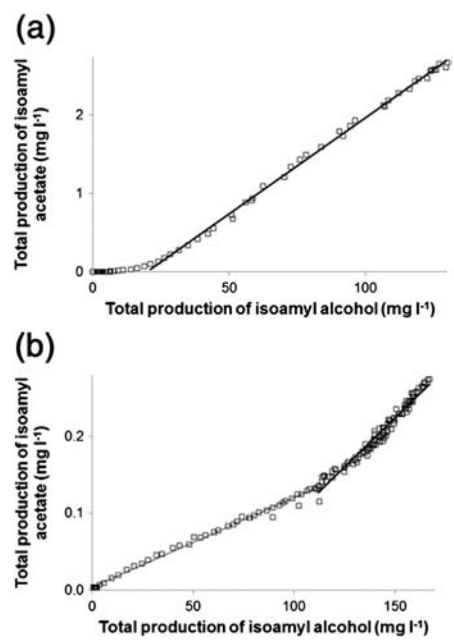

Fig. 6. (a) Changes in total isoamyl acetate production (□) as a function of total isoamyl alcohol production, for an initial assimilable nitrogen content of 410 mg N l^{-1}. For the linear phase, the regression coefficient was 0.998. (b) Changes in the total production of isoamyl acetate (□) as a function of total isoamyl alcohol production, for an initial assimilable nitrogen content of 70 mg N l^{-1}. For the two successive linear phases, the regression coefficients were 0.994 and 0.967, respectively. For the two graphs, the fermentation temperature was 18 °C.

For the three esters studied in particular, yields and total production over the entire fermentation process were not maximal in the same fermentation conditions (Tables 2 and 3a). For the acetate ester, total production was highest for the lowest temperature and the highest initial assimilable nitrogen content: 18 °C and 410 mg N l^{-1} of assimilable nitrogen (Table 2). The results for the two ethyl esters were more complex: total production was identical and maximal for two fermentation conditions: 18 °C and 410 mg N l^{-1}, and 24 °C and 410 mg N l^{-1}. The synthesis of these two groups of esters is catalyzed by different enzymes (Saerens et al., 2008; Verstrepen, Derdelinckx, et al., 2003): alcohol acetyl transferases (encoded by the ATF1 and ATF2 genes) for acetate esters and acyl-CoA:ethanol O-acyltransferases (encoded by the EEB1 and EHT1 genes) for ethyl esters. Thus, these enzymes and/or the other proteins involved in the metabolic pathways responsible for the production of these two types of esters appear to be regulated differently by temperature, but similarly by nitrogen content.

Finally, to obtain an overview of the dependence of total aroma compound production on temperature and initial nitrogen concentration, we described the production profiles of the six fermentative aromas, together with growth and CO_2 release, in terms of representative kinetic parameters: (1) start of synthesis (defined as the amount of sugar consumed at which the product concentration first exceeded 10% of its maximum concentration), (2) end of synthesis (defined as the amount of sugar consumed at which the product concentration reached 90% of the maximum concentration), (3) the amount of sugar consumed at which the production rate was maximal, (4) the amount of sugar consumed at which the specific production rate was maximal, (5) percentage production during the growth phase.

PCA was performed for detailed investigation of the effects of temperature and nitrogen availability on the chronology of compound synthesis (Fig. 7). The first two PCA axes accounted for 70.6% of the total variation. The principal parameters contributing to axis 1 were related to isoamyl acetate, cell count and propanol, whereas those contributing to axis 2 were mostly related to isobutanol. The distribution of individual fermentations on the basis of nitrogen content was similar for the first two axes of the PCA.

In Fig. 7, the fermentation conditions are grouped according to initial nitrogen content rather than temperature. The synthesis profiles of the six fermentative aromas studied here thus depended mostly on initial assimilable nitrogen content. Increases in temperature increased the rate of volatile compound synthesis without affecting the production profile of these compounds during the course of fermentation (Fig. 2a, c). However, there was an interaction between the effects of nitrogen and temperature on the regulation of yeast metabolism. Indeed, in Fig. 7, individual fermentations are not ranked in the same order for

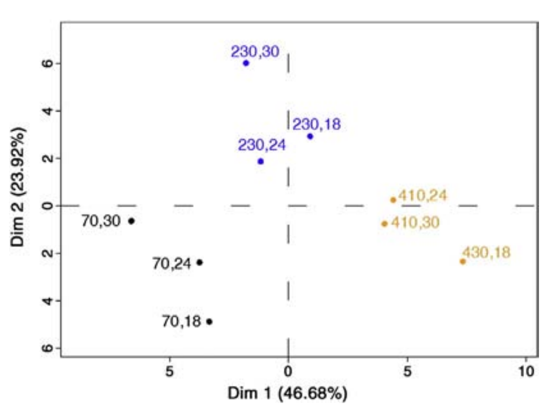

Fig. 7. Principal component analysis (PCA) of the kinetic parameters. The following kinetic parameters were determined for the six fermentative aromas, cell count and CO_2: (1) start of synthesis, (2) end of synthesis, (3) sugar consumption at the maximal production rate, (4) sugar consumption at the maximal specific production rate, (5) percentage production during the growth phase. Each fermentation is identified as X,Y, where X corresponds to the initial nitrogen concentration in mg N l^{-1} and Y is the fermentation temperature in °C.

the three nitrogen contents, according to axis 2. On the first axis of the PCA, for SM70 medium, the kinetic profiles were similar at 18 and 24 °C whereas, at high nitrogen contents (230 and 410 mg N l^{-1}), similarities were observed between 24 and 30 °C conditions.

3.2. Importance of the gas–liquid balance

In the previous sections, we present only the total production of fermentative aromas. Nevertheless, in published studies, the only data available are the liquid concentrations. We will show here that using the liquid content of fermentative aromas can lead to misinterpretations of the effects of fermentative parameters on the synthesis of aromas by yeast. Indeed, fermentative aromas are produced in the liquid phase by *S. cerevisiae*. However, these specific compounds are volatile and tend to evaporate, resulting in their partial loss in the exhaust gas. As a result, there are two distinct mechanisms accounting for the amounts of these volatile molecules accumulating in the liquid: biological production and physical evaporation.

Here, we compare data for total production with those for liquid concentration.

As shown by Morakul et al. (2013) and Mouret et al. (2013), losses in the exhaust gas were negligible for higher alcohols but much greater for esters (Table 2). Liquid concentration or total production can therefore be used indifferently to determine the quantities of higher alcohols effectively produced by *S. cerevisiae*. By contrast, total production must be used to determine the metabolic capacity for the synthesis of esters.

Final liquid concentrations of ethyl hexanoate and ethyl octanoate were systematically higher at 18 °C. However, the total production of these esters was identical at 18 and 24 °C for the three culture media (Table 2). Thus, the lower liquid concentration of ethyl esters at 18 °C than at 24 °C is due entirely to evaporation, rather than changes in yeast metabolism. Thus, analyses taking into account only liquid concentrations result in an overestimation of the effect of temperature on ester production. To our knowledge, this is the first time attempts have been made to differentiate between the physical and biological effects of temperature on the liquid accumulation of aromas in winemaking.

As for the final values (Table 2), calculations of the yield of production of fermentative aromas from sugars based on total productions or liquid concentrations gave similar values for higher alcohols but significantly different values for esters (Table 4). Indeed, the large losses observed for esters (Table 2) resulted in an underestimation of production yields from liquid concentrations (Table 4), as reported by Mouret et al. (2013). This underestimation was greater for the yields recorded for the second linear production phase (Table 4). Moreover, the second linear phase was shorter in analyses based on liquid content than in those based on total production, particularly for ethyl hexanoate (data not shown). For this specific compound, the second linear phase lasted from 20 to 170 g l^{-1} consumed sugar for calculations based on total synthesis and from 20 to 140 g l^{-1} consumed sugar for calculations based on liquid content. These findings reflect the increase in the percentage loss of esters during the course of the fermentation (Morakul et al., 2013; Mouret et al., 2012).

The yields for ester production from sugar in the two linear phases, calculated with total synthesis or liquid content, were used for PCA (Fig. 8a, b). The first two PCA axes accounted for 82% of the total variation for the liquid concentrations and 86% of the total variation for total production. In both PCAs, the main parameters contributing to axis 1 were the production yields of isoamyl acetate in phases 1 and 2, whereas the main parameters contributing to axis 2 were the production yields of ethyl hexanoate and ethyl octanoate in the first phase. The distributions of the nine fermentations differed, whether calculations were based on liquid content or total synthesis (Fig. 8a, b). For total production, the PCA clearly distinguished fermentations on the basis of their initial nitrogen content (Fig. 8a), indicating a key role for nitrogen in regulating the metabolism of ester synthesis. The discrimination was

Table 4
Underestimation of production yields of esters in calculations based on liquid concentrations.

Percentage underestimation for the production yield from sugar		SM70, 18 °C	SM70, 24 °C	SM70, 30 °C	SM230, 18 °C	SM230, 24 °C	SM230, 30 °C	SM410, 18 °C	SM410, 24 °C	SM410, 30 °C
Isoamyl acetate	First phase	21	31	43	7	11	16	9	15	23
	Second phase				18	28	41	14	23	36
Ethyl hexanoate	First phase	13	17	16	12	14	22	18	13	28
	Second phase	31	36	33	33	44	48	21	24	57
Ethyl octanoate	First phase	14	31	19	21	29	38	25	21	41
	Second phase	49	42	41	32	41	51	29	42	55
Percentage underestimation for the production yield from isoamyl alcohol		SM70, 18 °C	SM70, 24 °C	SM70, 30 °C	SM230, 18 °C	SM230, 24 °C	SM230, 30 °C	SM410, 18 °C	SM410, 24 °C	SM410, 30 °C
Isoamyl acetate	First phase	17	23	33	18	26	39	14	22	35
	Second phase	22	30	45						

less clear-cut when we used liquid concentrations (Fig. 8b), particularly for media with high nitrogen contents (250 and 430 mg N l^{-1}). This observation confirms the need for gas–liquid balances for a good understanding of ester production: at 30 °C, a change in yeast metabolism resulted in lower levels of ester synthesis, but the low liquid concentrations observed mostly reflected the effects of evaporation.

4. Conclusion

In this work, we studied the impact of initial assimilable nitrogen concentration and temperature on the production kinetics of the principal fermentative aromas. Gas–liquid balances were also determined, to differentiate between the biological and physical consequences of differences in temperature. We found that the impact of temperature on ester synthesis was overestimated if only liquid content was considered.

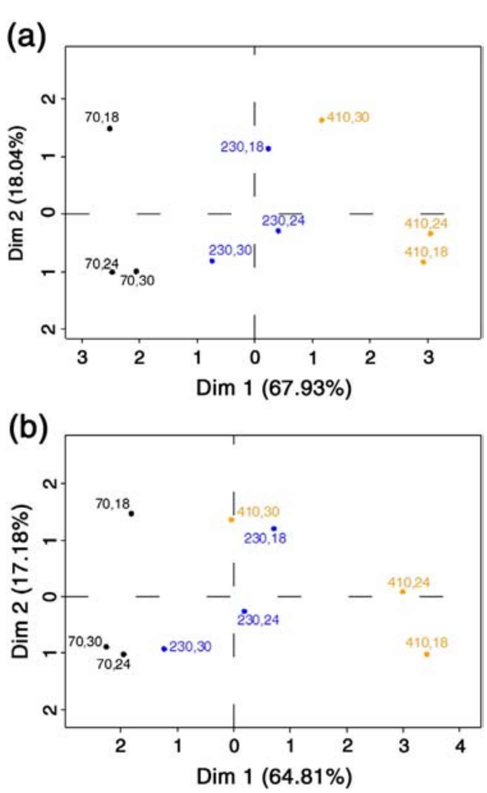

Fig. 8. PCA of the production yields for the three esters (isoamyl acetate, ethyl hexanoate and ethyl octanoate) from sugars, calculated either from total productions (a) or from liquid concentrations (b). Each fermentation is identified as X,Y, where X is the initial nitrogen concentration in mg N l^{-1} and Y is the fermentation temperature in °C.

Assimilable nitrogen appeared to be the major fermentation parameter affecting the kinetics of synthesis for higher alcohols and esters, but temperature also had an effect, both on its own and in interaction with nitrogen. Original correlations between the production of fermentative aromas and sugar consumption were established for all the compounds studied. Propanol was identified as a quantitative marker of initial nitrogen content and a metabolic indicator of the availability of intracellular nitrogen.

Additional experiments in other strains and in media with different compositions (especially natural musts) are required, but these results provide preliminary data and insight, making it possible to model the dynamics of synthesis for fermentative aromas. These findings will also facilitate the development of innovative strategies for controlling the production of fermentative aromas by managing assimilable nitrogen content and temperature during alcoholic fermentation.

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